

WorldScape Policy 2.0: Empowering Steerable World Action Modeling with Reasoning-Augmented Memory

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Abstract. World Action Models (WAMs) offer a promising paradigm for robotic manipulation by jointly modeling visual state transitions and robot actions. However, existing WAMs are constrained by limited temporal context, coarse episode-level language supervision, and predominantly text-only conditioning, which hinder task-progress tracking and fine-grained language-video-action grounding while limiting visual-context reasoning and cross-embodiment transfer. In this paper, we introduce **WorldScape Policy 2.0**, a controllable WAM with reasoning-augmented long short-term memory. Its causal short-term visual memory supplies recent observations as DiT prefill to preserve local interaction dynamics, while its long short-term event memory organizes historical VLM outputs into global-history, local-active, and event-boundary representations for progress-aware retrieval. The retrieved history augments perception and autoregressively generated planning tokens, yielding an implicit subgoal condition for autonomous planning; semantic forcing further transfers event-level instruction semantics into this latent planning pathway. To establish fine-grained multimodal controllability, we construct **ManipEvent-5M**, an event-grounded embodied pretraining dataset containing nearly 5 million event segments with aligned action trajectories, episode-level task instructions, segment-level subtask captions, goal images, and video demonstrations. These designs provide a unified interface for autonomous planning from high-level instructions and controllable execution from fine-grained text, goal-image, or video-context prompts. Experiments in both simulation and real-world platforms demonstrate superior capabilities in long-horizon autonomous planning, fine-grained instruction following and in-context adaptation.

Date: July 20, 2026

Webpage: <https://manifoldai-research.github.io/WorldScape-Policy/>

1 Introduction

Robotic learning is shifting from task-specific policies toward generalist embodied models that can interpret task intent, anticipate action consequences, and adapt to new tasks from contextual cues. Vision-Language-Action (VLA) models have demonstrated the promise of mapping language and perception to executable robot actions, but they typically emphasize direct policy prediction rather than how the visual world evolves under those actions. World Action Models (WAMs) provide a complementary paradigm by jointly modeling visual state transitions and robot actions. This coupling provides a natural basis for action planning and opens a viable pathway toward context-conditioned adaptation. However, transforming a WAM from a short-horizon predictive model into a controllable long-horizon policy requires addressing three interrelated questions: what historical information should be retained, how atomic actions should be grounded through fine-grained supervision, and how task intent can be expressed beyond language alone.

Despite remarkable progress, current WAMs remain limited along these three dimensions, as illustrated in Fig. 1. **Limited progress-aware memory.** Most existing methods condition world-action modeling on either the current observation [1–4] or a short history window [5, 6]. Such limited context obscures global task progress when different stages share similar visual observations. For example, a tabletop may appear nearly identical before and after an intermediate subgoal, although the correct next action depends on what has already been completed. MemoryWAM [7] expands access to the full visual history through an efficient hybrid memory mechanism. However, retrieving past visual evidence alone does not explicitly organize history around task progress or convert it into a latent subgoal condition for autonomous planning.

Coarse language-video-action grounding. In many embodied datasets, different segments of an episode inherit the

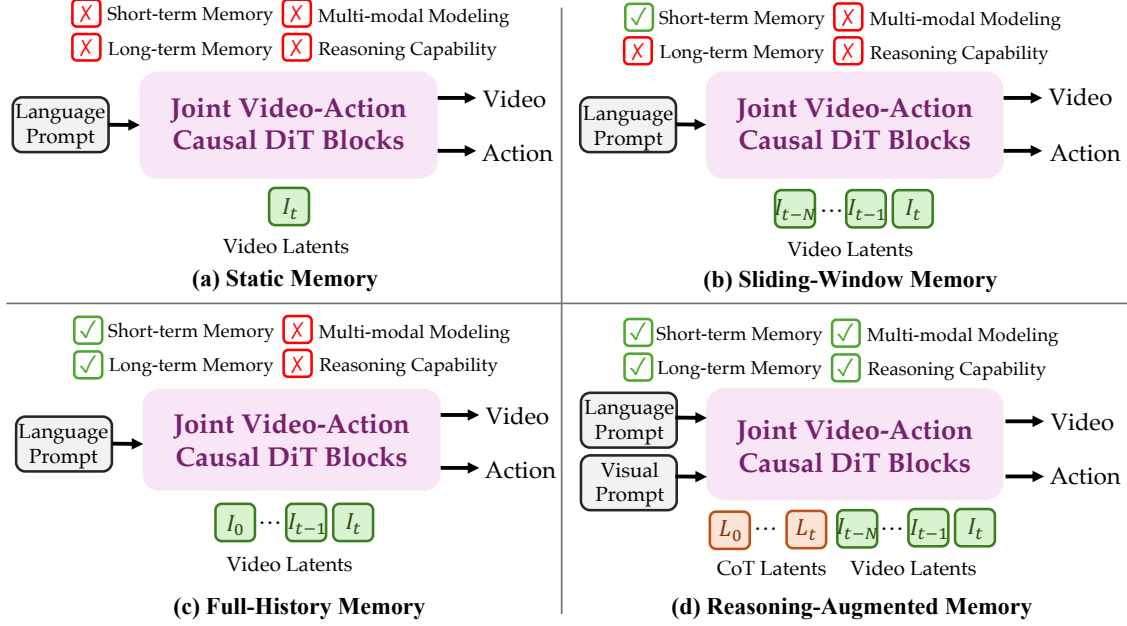


Figure 1: Comparison of different WAM paradigms. Existing models use (a) static observations, (b) short-term visual history, or (c) full-history visual memory without progress-aware reasoning or multimodal control. (d) WorldScape Policy 2.0 introduces multimodal controllability and reasoning-augmented long short-term memory, enabling interactive video-action modeling for long-horizon robotic manipulation.

same task-level instruction, providing weak linguistic supervision for diverse atomic actions. Models can therefore rely on visual shortcuts instead of learning how an event-level intent aligns with a specific state transition and action chunk. Consequently, a model may generate plausible video-action trajectories while remaining insensitive or brittle to fine-grained changes in the requested atomic action.

Restricted interaction interfaces. Existing WAMs are also largely controlled through text, whereas users may express intent more naturally through a goal image, a demonstration video, or an example from another embodiment. Without native visual prompting, a WAM cannot fully exploit target-state evidence, object correspondences, or latent task rules, limiting both visual-context reasoning and cross-embodiment transfer.

Our key insight is that long-horizon controllability requires two complementary forms of temporal context: semantic event memory for reasoning about task progress and frame-level visual memory for preserving local interaction dynamics. Based on this insight, we propose **WorldScape Policy 2.0**, a controllable WAM with reasoning-augmented long short-term memory. Its VLM branch organizes historical chunks into global-history, local-active, and event-boundary memory views, then retrieves relevant evidence to augment perception and autoregressive planning tokens of current observation. In parallel, its causal DiT retains recent observations as short-term visual prefill and optional visual prompts as persistent context. This division enables event memory to infer the next subgoal while visual memory preserves evolving interaction dynamics.

To provide scalable supervision, we construct **ManipEvent-5M**, an event-based multimodal dataset containing nearly 5 million segments aligned with action trajectories, episode-level task instructions, event-level subtask captions, goal images, and video demonstrations. Our three-stage curriculum first establishes fine-grained controllability and causal visual memory through event-grounded pretraining, then introduces the event memory branch which transfers event semantics into autonomous latent planning through semantic forcing, and finally adapts the model to downstream embodiments and various interaction modes. The resulting joint video-action model supports autonomous planning from high-level instructions, direct control from fine-grained captions, and in-context adaptation from visual prompts.

Our main contributions are summarized as follows:

- We propose **WorldScape Policy 2.0**, a controllable WAM that couples progress-aware long short-term event

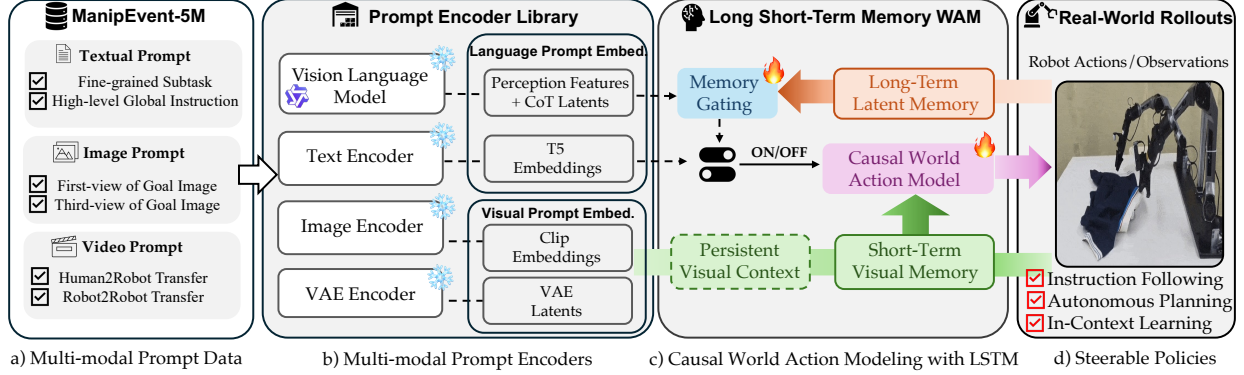


Figure 2: Overview of the WorldScape Policy 2.0 framework. Built on the event-based ManiEvent-5M pretraining data, WorldScape Policy 2.0 encodes observations together with interchangeable text, goal-image, and video-demonstration prompts into a unified causal video-action modeling pipeline. The model couples short-term visual transitions with long-term reasoning-augmented event memory, enabling implicit subgoal planning and joint video-action prediction. This design supports both autonomous planning from coarse task instructions and controllable manipulation from fine-grained subtask captions or visual prompts.

memory with causal short-term visual memory for implicit subgoal planning in long-horizon manipulation.

- We introduce **ManiEvent-5M**, an event-grounded embodied pretraining dataset containing nearly 5 million segments with aligned action trajectories, hierarchical text instructions, goal images, and video demonstrations, enabling fine-grained multimodal WAM interactions.
- We develop a three-stage learning strategy that transfers event-level semantics from controllable pretraining to autonomous planning through semantic forcing and supports a unified interface for high-level planning, atomic instruction following, and visual-prompted adaptation. Experiments on simulation benchmarks and a dual-arm PiPER platform evaluate these capabilities and the contribution of each design component.

2 Related Work

2.1 World Action Models for Robotic Manipulation

World Action Models (WAMs) leverage visual dynamics for robot action planning, offering a dynamics-centric alternative to reactive observation-to-action policies. Early world models and video-prediction policies use future observations, visual goals, or latent states as intermediate planning representations [8–15]. V-JEPA 2-AC further demonstrates that an action-conditioned latent world model, post-trained from large-scale video representations with limited robot data, can support image-goal planning [16]. Complementary approaches use world models as scalable embodied-data engines, as in GigaWorld-0 [17], or directly adapt pretrained video models into policies, as in Cosmos Policy [18], which represents actions, future states, and values within the video latent space.

Recent WAMs move toward unified video-action modeling, showing that action-centered or causal video-action models can serve as zero-shot robotic policies [1, 2, 4, 6]. LingBot-VA [5] emphasizes causal autoregressive video-action modeling with persistent KV-cache, while MemoryWAM [7] improves memory efficiency with recent frames, event-boundary anchors, and gist tokens. Beyond holistic video or global-latent prediction, OA-WAM introduces persistent object-addressable slots and jointly predicts object-centric world states and flow-matching action chunks [19]. However, prior WAMs often retrofit text-guided video backbones, use video prediction mainly as training-time supervision, or lack explicit task-progress reasoning [1, 4]. **WorldScape Policy 2.0 addresses these gaps through native controllable WAM pretraining and reasoning-augmented long short-term memory.**

2.2 Long-Term Memory and Long-Horizon Planning

Long-horizon manipulation requires models to remember completed subgoals and decide what remains to be done. End-to-end approaches such as Long-VLA [20] introduce phase-aware modeling of movement and interaction to improve skill chaining without an explicit planner. Other VLAs expose intermediate planning representations: $\pi_{0.5}$ [21] predicts high-level subtasks before action generation, while $\pi_{0.7}$ [22] uses subtask language, metadata, history, and visual subgoals for long-horizon and language-coached execution. VLA-OS [23] systematically studies language, visual, and hierarchical planning representations, highlighting the benefit of visually grounded intermediate plans.

Memory-oriented methods such as MEM [24] and MemoryVLA [25] retrieve and fuse multi-scale embodied or perceptual-cognitive memory for temporally aware action prediction. LoHo-Manip [26] maintains an explicit done-and-remaining subtask plan together with an updated visual trace under receding-horizon control. Dual-system methods, including Goal2Skill [27] and DSWAM [28], similarly decouple high-level planning from low-level execution by passing subgoals, visual traces, or executable instructions to VLA / WAM executors. NovaPlan [29] combines a closed-loop VLM planner with generated video references and geometrically grounded execution for zero-shot long-horizon manipulation. These works highlight the value of memory and subtask reasoning, but typically rely on VLA policies, external planners, or explicit intermediate plans. **WorldScape Policy 2.0 instead integrates recent visual dynamics, event-level progress memory, and latent subgoal reasoning within a controllable WAM.**

2.3 Fine-Grained Prompts for Steerable Policies

Fine-grained language prompts provide an important control interface for steerable manipulation, since coarse trajectory-level instructions cannot specify diverse atomic actions, object states, execution styles, and intermediate subgoals. At the generalist-model level, Qwen-VLA [30] unifies continuous action and trajectory generation across tasks, environments, and robot embodiments through embodiment-aware prompting, while retaining fine-grained control within the same policy. FineVLA [31] instead isolates language granularity as a supervision axis, aligning goal-level and process-level instructions with robot trajectories to improve factor-level steerability. Related works introduce dense control or grounding signals via pixel-level visual prompts [32], 3D spatial action representations [33], reconstruction-based visual grounding [34], and auxiliary spatial co-training [35].

Visual prompting provides a complementary interface for specifying where and how to act. TraceVLA [36] renders state-action trajectories as visual traces to improve spatiotemporal grounding, whereas CoT-VLA [37] autoregressively predicts future subgoal images as visual reasoning steps before action generation. Human-demonstration-conditioned policies [38] further use a single human video as a task prompt and align human and robot representations for cross-embodiment transfer. Explicit subtask decomposition can also provide fine-grained executable instructions to WAM executors [28]. **WorldScape Policy 2.0 unifies these complementary directions through event-grounded WAM pretraining, aligning fine-grained subtask captions, goal images, and video demonstrations with visual state transitions and atomic action trajectories in ManipEvent-5M.**

3 Method

3.1 Overview and Problem Formulation

WorldScape Policy 2.0 is a controllable World Action Model designed to support autonomous planning and interactive manipulation within a shared video-action backbone. Let o_t denote the multi-view observation at time t , a_t the corresponding robot action, and z_t the VAE-encoded visual latent of o_t . We distinguish two language interaction modes, $\mu \in \{\text{auto}, \text{fine}\}$, whose language inputs are an episode-level instruction y and an event-level instruction c_t , respectively. The corresponding prompt set is $\mathcal{P}_t^\mu = \{p_{\text{lang}}^\mu, p_{\text{goal}}, p_{\text{video}}\}$, where $p_{\text{lang}}^{\text{auto}} = y$ and $p_{\text{lang}}^{\text{fine}} = c_t$. Historical context is summarized by $\mathcal{M}_t = (Q_t, \mathcal{Z}_t^{\text{vis}})$, comprising the event-memory queue Q_t and the recent visual-latent buffer $\mathcal{Z}_t^{\text{vis}}$. The event queue is activated for autonomous planning after Stage 2 and is empty when event memory is disabled, whereas the visual buffer is used throughout all training stages. The model predicts an action chunk and future visual latents as:

$$p_\theta(a_{t:t+H}, z_{t+1:t+H} \mid o_t, a_{<t}, \mathcal{P}_t^\mu, \mathcal{M}_t). \quad (1)$$

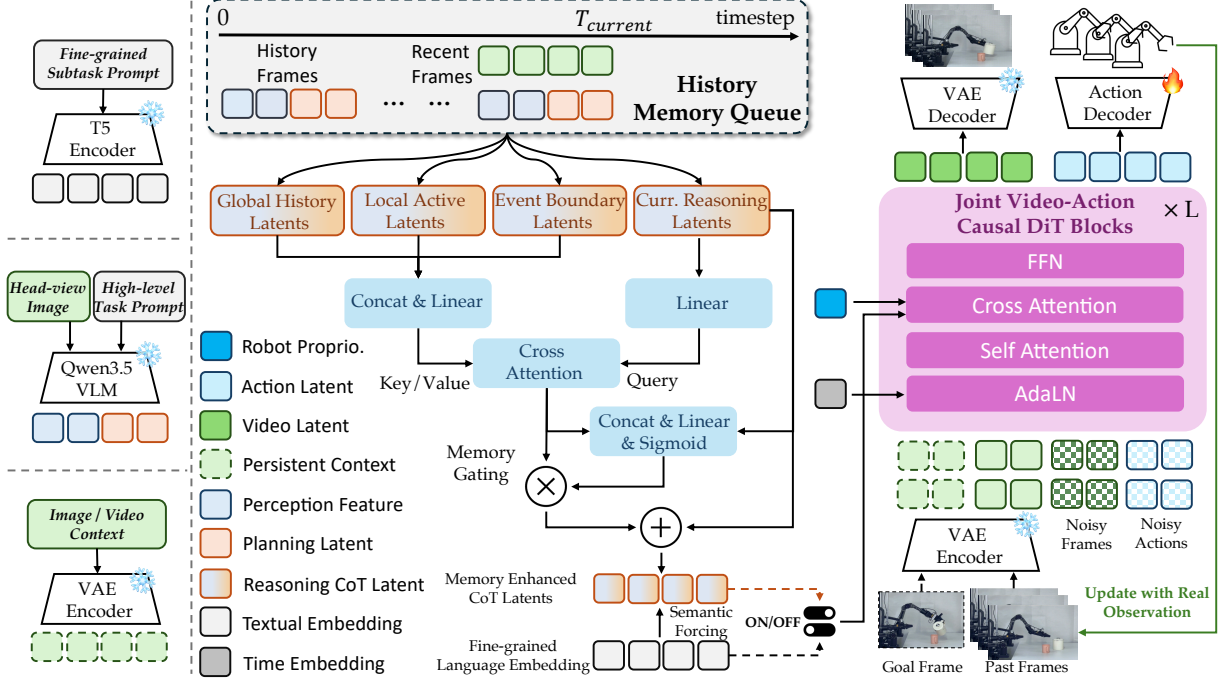


Figure 3: Illustration of Reasoning-Augmented Long Short-Term Memory. Historical VLM outputs form global-history, local-active, and event-boundary memories, which are retrieved and gated to augment the current reasoning tokens; semantic forcing aligns them with subtask semantics. Recent observations and optional visual prompts enter the DiT as causal visual prefill, while memory-enhanced VLM tokens or T5 embeddings condition joint video-action flow prediction according to the interaction mode.

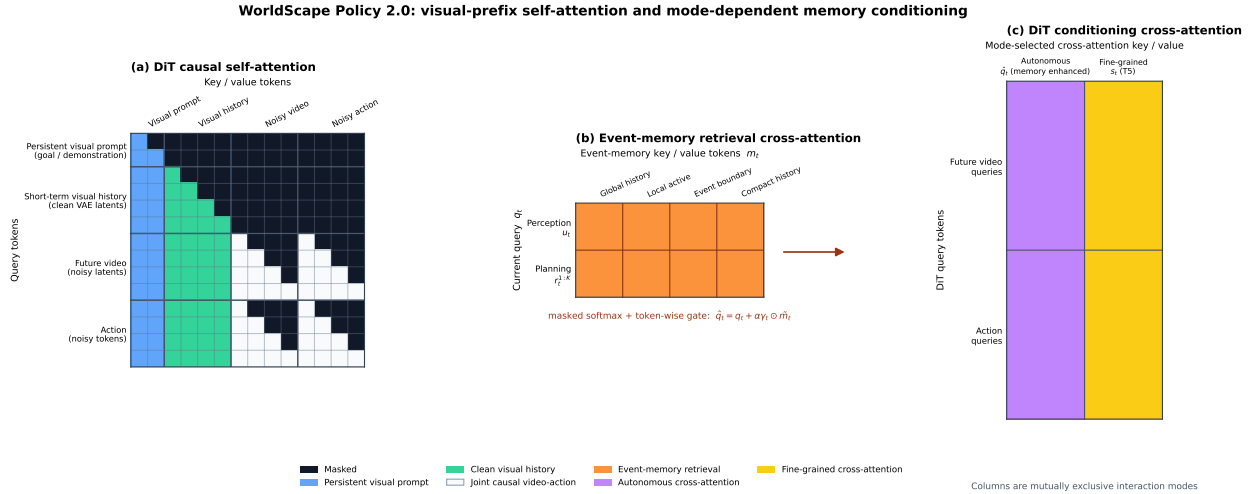


Figure 4: Attention pattern of WorldScape Policy 2.0. In the DiT self-attention mask, persistent visual prompts and clean short-term visual history are placed before the noisy future video and action chunks, so generated tokens can reuse goal, demonstration, and recent observation context without accessing future clean targets. In autonomous planning, event memory is injected through a separate cross-attention path from current reasoning tokens to retrieved global-history, local-active, event-boundary, and compact-history memory tokens; the resulting memory-enhanced VLM tokens condition the DiT. Fine-grained instruction following instead uses T5 tokens as the DiT cross-attention condition.

(1) *Unified action representation.* For robot embodiment e , each per-arm action is represented as $[p_t^{(e)}; r_t^{(e)}; g_t^{(e)}] \in \mathbb{R}^{10}$, where $p_t^{(e)} \in \mathbb{R}^3$ is the Cartesian position, $r_t^{(e)} \in \mathbb{R}^6$ is a continuous 6D rotation representation, and $g_t^{(e)} \in \mathbb{R}$ is the one-DoF gripper command. For a dual-arm embodiment, the two per-arm vectors are concatenated to form the raw

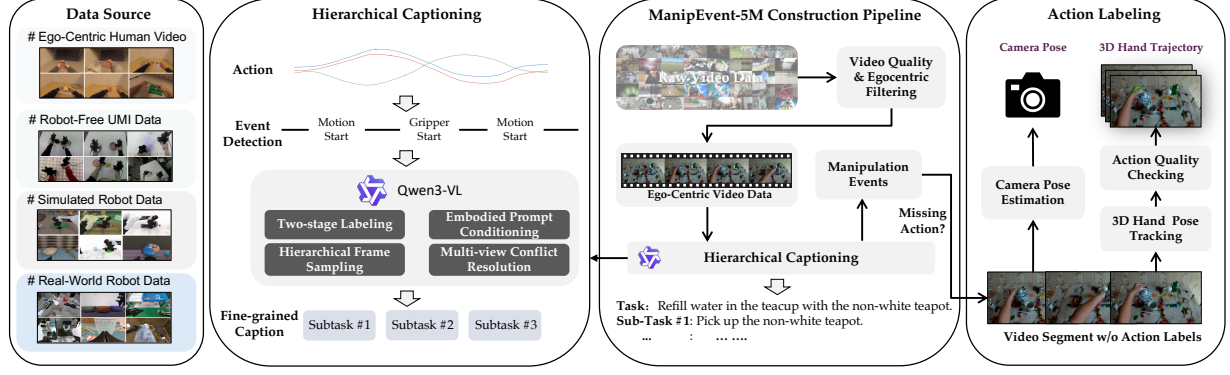


Figure 5: Construction pipeline of ManipEvent-5M. We aggregate heterogeneous manipulation sources, including human-arm ego videos, robot-free UMI data, simulated trajectories, and real-robot demonstrations, and convert them into event-level training samples. Each trajectory is segmented into ordered atomic subgoals and annotated with paired action trajectories, episode-level global instructions, segment-level fine-grained captions, goal-image prompts, and video-context prompts. This event-based multimodal schema supports the event-grounded WAM interface in Fig. 2, enabling fine-grained language grounding, goal-conditioned policy learning, demonstration-conditioned in-context learning, and long-horizon memory mid-training.

action $a_t^{(e)} \in \mathbb{R}^{20}$.

(2) *Embodiment-specific action adapters.* Each embodiment has a dedicated action encoder $E_a^{(e)}$ and action decoder $D_a^{(e)}$:

$$\tilde{a}_\rho^{(e)} = E_a^{(e)}(a_\rho^{(e)}), \quad \hat{v}_\theta^{a, (e)} = D_a^{(e)}(h_\theta^{a, (e)}), \quad (2)$$

where $E_a^{(e)}$ projects a flow-perturbed raw action into the shared DiT token space and $D_a^{(e)}$ maps the resulting action hidden states back to a velocity field in the raw action space. The causal video-action DiT is shared across embodiments, while these adapters absorb embodiment-specific kinematics, action scales, and control conventions. Importantly, the adapters only provide the input and output interfaces: flow matching and flow integration are both performed on the raw action rather than on $\tilde{a}_\rho^{(e)}$.

As shown in Fig. 2, language and visual prompts follow distinct conditioning paths. Fine-grained instructions are encoded by T5, high-level instructions are interpreted jointly with the current observation by the VLM reasoning branch, and visual prompts are encoded as persistent VAE prefill. The resulting mode-specific conditions are integrated with event and visual memory before the DiT jointly predicts future video and action chunks.

3.2 Event-Grounded World Action Modeling

Existing WAMs are often trained with coarse episode-level instructions or a single interaction modality, providing weak supervision for the diverse state changes and atomic actions inside long manipulation episodes. WorldScape Policy 2.0 instead grounds WAM learning at the event level, where each segment aligns fine-grained subtask semantics with visual transitions and paired robot actions. This event-level structure supports two language interaction modes and complementary visual prompting. A fine-grained subtask instruction is encoded by T5 and directly controls the WAM, whereas a high-level task instruction is interpreted by the VLM branch to support autonomous subgoal planning. Goal images and video demonstrations provide additional persistent visual context. We write the directly encoded prompt tokens as

$$s_t = E_{T5}(c_t), \quad g_t = E_{VAE}(p_{\text{goal}}), \quad v_t = E_{VAE}(p_{\text{video}}), \quad (3)$$

where c_t is a fine-grained subtask instruction, s_t is its T5 embedding, and g_t and v_t are clean VAE latents of goal images and video demonstrations. The persistent visual condition is selected according to the available visual prompt:

$$p_t^{\text{vis}} = \begin{cases} g_t, & \text{goal-image prompting,} \\ v_t, & \text{video-demonstration prompting,} \\ [g_t; v_t], & \text{both prompts available,} \\ \emptyset, & \text{otherwise.} \end{cases} \quad (4)$$

The language condition is likewise selected by μ , rather than concatenating the T5 and VLM branches into a single input.

Mode-dependent language grounding. In fine-grained instruction following, the segment-level caption c_t explicitly specifies the atomic action or intermediate subgoal. Its T5 embedding s_t is directly injected into the DiT through cross-attention, enabling precise and steerable action execution. In autonomous planning, the robot receives only an episode-level high-level instruction y . The VLM jointly interprets y , the current observation, and historical event memory to infer the active subgoal; the resulting memory-enhanced VLM tokens, rather than a T5 embedding of y , condition the DiT. Event-level captions supervise this latent planning pathway through semantic forcing during training.

Goal-conditioned policy learning. Goal image prompts specify the desired target state visually. WorldScape Policy 2.0 encodes the goal image as g_t and retains it in the causal visual prefill, allowing flow-perturbed future chunks to attend to the desired state throughout generation. The model is thus trained to infer an executable action path from the current observation to the visual goal.

Demonstration-conditioned in-context adaptation. Video context prompts introduce cross-embodiment adaptation tasks, where the context may come from human-to-robot demonstrations, robot-to-robot demonstrations, or previous contextual episodes with different embodiments and viewpoints. WorldScape Policy 2.0 represents these demonstrations as v_t and retains them as persistent visual prefill context alongside recent observations. This training mode encourages the model to infer object correspondences, transferable action patterns, and latent task rules from visual demonstrations, directly matching the video-context annotations provided by ManipEvent-5M.

3.3 Reasoning-Augmented Long Short-Term Memory

Long-horizon manipulation requires memory at two complementary representation levels. At the event level, the model must retain semantic evidence about completed subtasks and retrieve history relevant to the current task stage. At the visual level, it must preserve recent frame-level dynamics, such as contact changes and object motion, for temporally coherent video-action prediction. As illustrated in Fig. 3, WorldScape Policy 2.0 therefore combines a *long short-term event memory* in the VLM reasoning branch with a *short-term visual memory* in the causal DiT branch. The former performs progress-aware retrieval over historical chunk representations, whereas the latter directly supplies recent visual latents as causal prefill context. Optional goal images and video demonstrations are further retained as persistent visual context throughout the rollout.

Latent reasoning formulation. For each action chunk under autonomous planning, the VLM jointly processes the current head-view observation and episode-level instruction. Let $\{h_t^\ell\}$ denote its layer-wise hidden states. The perception branch extracts a visual-language representation

$$u_t = A_{\text{perc}}(\{h_t^\ell\}), \quad (5)$$

where A_{perc} is the configured feature extractor, implemented using either the final VLM hidden sequence or a layer-wise QFormer that maps selected VLM layers into a fixed number of compact perception tokens.

In parallel, the planning branch uses the same head-view observation together with a task-conditioned planning prompt $\tilde{y}_t = T_{\text{plan}}(y)$. We instantiate T_{plan} with the following fixed template, where {task} is replaced by the episode-level instruction y :

“You are a robot planner. Instructions: {task}. Given the current high-level task instruction and current head-view observation, predict the next atomic action subtask for the next second.”

Conditioned on this prompt and the current observation, the planning branch autoregressively generates K discrete planning tokens using greedy decoding:

$$w_t^k = \arg \max_w p_{\text{VLM}}(w \mid o_t, \tilde{y}_t, w_t^{<k}), \quad k = 1, \dots, K. \quad (6)$$

We then perform a full-sequence VLM forward pass over the input context and generated sequence, and take the final-layer hidden states at the generated token positions as latent planning tokens:

$$r_t^k = [H_{\text{VLM}}^L(o_t, \tilde{y}_t, w_t^{1:K})]_{\text{pos}(w_t^k)}, \quad k = 1, \dots, K. \quad (7)$$

Unlike the perception tokens, which encode the observed visual-language context, these planning features summarize the VLM’s autoregressively inferred task continuation. We concatenate the two token groups to form the current reasoning latent

$$q_t = [u_t; r_t^{1:K}], \quad (8)$$

which jointly represents the current state and the inferred next subgoal. Historical observations are processed using the same perception and planning branches, ensuring that every memory entry has the same two-part token structure.

Long short-term event memory construction. The event-memory branch maintains the queue $\mathcal{Q}_t = \{q_1, \dots, q_{t-1}\}$ of historical chunk-level VLM outputs introduced above. Each q_j contains the perception tokens and latent planning tokens associated with chunk j . To control the cost of full-history retrieval, we compress the perception tokens of every historical chunk into a fixed number of learned gist tokens through attention pooling, while retaining its planning tokens. Concatenating the resulting tokens gives a compact full-history bank H_t . We then explicitly construct three complementary memory views. Global-history latents summarize the complete trajectory, local-active latents represent the currently active event, and event-boundary latents sparsely preserve major semantic transitions:

$$\begin{aligned} m_t^{\text{gh}} &= P_{\text{gh}}(\text{Expand}(F_{\text{gh}}([e_y; \text{Mean}(H_t)]))), \\ m_t^{\text{la}} &= P_{\text{la}}([q_j \mid j \in \mathcal{I}_t^{\text{la}}]), \quad \mathcal{I}_t^{\text{la}} = \{\max(1, t - S_e), \dots, t - 1\}, \\ m_t^{\text{eb}} &= P_{\text{eb}}([q_j \mid j \in \mathcal{B}_t]), \end{aligned} \quad (9)$$

Here, e_y denotes the embedding of the high-level task instruction, F_{gh} fuses task intent with the mean-pooled history, and Expand produces a fixed number of global-history slots. The global-history latents m_t^{gh} therefore provide a task-conditioned summary of completed subtasks and overall trajectory progress. The local-active latents m_t^{la} model the currently active event and recent semantic transitions over the latest S_e chunks. In implementation, these recent chunks are retained as full-token anchor frames rather than further compressed, preserving fine-grained perception and planning evidence for the ongoing subtask.

Event-boundary chunks are selected directly from latent changes in the historical VLM outputs. Let $\bar{q}_j = \text{Mean}(q_j)$ denote the token-averaged representation of chunk j . We compute

$$d_j = 1 - \cos(\bar{q}_j, \bar{q}_{j-1}), \quad \mathcal{B}_t = \text{TopK}_{\Delta}(\{d_j\}_{j=2}^{t-1}; S_b), \quad (10)$$

where d_j measures the semantic change between consecutive chunks. TopK_{Δ} greedily selects at most S_b highest-change indices while enforcing a minimum temporal separation Δ between selected indices. The resulting event-boundary latents sparsely retain states at which a subtask likely starts, terminates, or changes, without requiring explicit online boundary annotations. As with local-active memory, the selected chunks are stored as full-token anchor frames, allowing the model to revisit detailed perception and planning evidence at these critical transitions rather than only their pooled boundary scores. Consequently, event memory combines long-term task-progress context from global-history and event-boundary latents with short-term semantic context from local-active latents. The anchor-frame implementation preserves local detail within this semantic memory organization, which remains distinct from the frame-level visual memory described below.

Event-memory retrieval and gated fusion. Following the implementation, the three specialized memory views are concatenated with the compact full-history bank, ensuring that selection does not discard non-boundary evidence. A shared value projection forms the memory bank

$$m_t = P_v([m_t^{\text{gh}}; m_t^{\text{la}}; m_t^{\text{eb}}; H_t]). \quad (11)$$

The current reasoning latent serves as the query and performs scaled dot-product cross-attention over the valid memory tokens:

$$\tilde{m}_t = \text{softmax}\left(\frac{P_q(q_t)m_t^{\top}}{\sqrt{D}} + M_t\right)m_t, \quad (12)$$

where M_t masks padded history and boundary slots and D is the token dimension. Because historical evidence is not equally useful at every step, a learned gate controls how much retrieved memory is injected into each current token:

$$\gamma_t = \sigma(W_g[q_t; \tilde{m}_t] + b_g), \quad \hat{q}_t = q_t + \alpha \gamma_t \odot \tilde{m}_t, \quad (13)$$

where γ_t is a token-wise memory-importance gate, α is a residual scaling factor, and $\hat{q}_t$ is the memory-enhanced reasoning latent. This retrieval mechanism allows the model to selectively use task-level progress, recent event context, sparse transition evidence, and the compressed full history.

Short-term visual memory and persistent visual context. In parallel to event-memory retrieval, the causal DiT models recent frame-level dynamics through visual prefilling. We define the recent visual-latent buffer as $\mathcal{Z}_t^{\text{vis}} = z_{t-S_v:t}^{\text{obs}}$, containing the clean VAE latents of the current observation and the most recent S_v visual chunks. These latents precede the flow-interpolated future video latents and embodiment-encoded action tokens in the DiT sequence:

$$h_t^{\text{DiT},(e)} = [p_t^{\text{vis}}; \mathcal{Z}_t^{\text{vis}}; z_{t+1:t+H}^{\rho}; E_a^{(e)}(a_{t:t+H}^{\rho,(e)})], \quad (14)$$

where $\mathcal{Z}_t^{\text{vis}}$ is updated online after each executed action chunk and $E_a^{(e)}$ is selected according to the current embodiment. As summarized in Fig. 4, a causal visual-attention mask allows future video-action tokens to attend to persistent visual prompts and available clean history while preventing information leakage from future chunks. This pathway supplies local motion and interaction continuity through DiT self-attention, rather than first compressing recent frames into the VLM event memory.

The optional prompt latents p_t^{vis} remain fixed across rollout steps, unlike the sliding recent-frame buffer, allowing the model to continuously reference the desired goal state or demonstrated trajectory. Both recent and persistent visual latents enter the DiT through causal self-attention, while the cross-attention condition is selected by interaction mode: T5 embeddings are used for fine-grained instruction following, whereas memory-enhanced VLM tokens are used for autonomous planning. This separation preserves the respective roles of semantic event retrieval, local visual dynamics, and multimodal task conditioning.

3.4 Implicit Subgoal Latent Planning

WorldScape Policy 2.0 uses event-level subtask captions in two different ways. In fine-grained instruction following, the T5 embedding $s_t = E_{\text{T5}}(c_t)$ acts as the direct language condition. In autonomous planning, the caption is used only as training-time semantic supervision for the VLM planning branch, which produces $\hat{q}_t$ from perception, autoregressive planning tokens, and retrieved historical memory.

Subgoal semantic forcing. During training, s_t provides a semantic target for the autonomous-planning latent. Because s_t and $\hat{q}_t$ are token sequences of potentially different lengths, we first obtain fixed-dimensional summaries $\bar{s}_t = \text{Pool}(s_t)$ and $\bar{q}_t = \text{Pool}(\hat{q}_t)$. We then apply *semantic forcing* in a shared projection space:

$$\mathcal{L}_{\text{sem}} = 1 - \frac{\phi_s(\bar{s}_t)^\top \phi_q(\bar{q}_t)}{\|\phi_s(\bar{s}_t)\|_2 \|\phi_q(\bar{q}_t)\|_2}, \quad (15)$$

where ϕ_s and ϕ_q project language and reasoning summaries into the same semantic space. This alignment transfers explicit subtask semantics into the latent planning representation, encouraging it to encode atomic-action intent and task progress. During autonomous inference, the fine-grained caption and its T5 embedding are absent; the model receives only the high-level instruction and relies on $\hat{q}_t$ to infer the active subgoal from the current observation and event memory.

Mode-dependent video-action prediction. The language-side cross-attention condition is selected according to the requested interaction mode:

$$\ell_t = \begin{cases} s_t, & \text{fine-grained instruction following,} \\ \hat{q}_t, & \text{autonomous planning,} \end{cases} \quad (16)$$

where s_t provides explicit atomic-action control, while $\hat{q}_t$ provides an implicitly inferred and progress-aware subgoal condition. Let ρ denote the flow timestep embedded through AdaLN. The shared causal DiT produces action and video hidden states, after which the embodiment-specific action decoder and shared video output head predict their respective velocity fields:

$$\begin{aligned} (h_\theta^{a,(e)}, h_\theta^z) &= F_\theta(h_t^{\text{DiT},(e)}, \ell_t, \rho), \\ \hat{v}_\theta^{a,(e)} &= D_a^{(e)}(h_\theta^{a,(e)}), \quad \hat{v}_\theta^z = D_z(h_\theta^z), \end{aligned} \quad (17)$$

where F_θ denotes the shared causal video-action DiT, D_z is the video output head, and visual prompts, when available, are already included in $h_t^{\text{DiT},(e)}$ as persistent prefill context. At inference, the video field is integrated in VAE-latent

Table 1: Statistics of WorldScape Policy 2.0 pretraining data in ManipEvent-5M. The number of segments is reported for datasets with event-level decomposition. “Avg. Seg. / Episode” is the number of event segments divided by the number of episodes, and “Single-Seg. Ratio” is the proportion of episodes without further event decomposition.

Data Type	Dataset Name	Frames (M)	Duration (H)	Episodes (K)	Segments (M)	Avg. Seg. / Episode	Single-Seg. Ratio (%)	Training Ratio (%)
Real-World Robot Data	Self-Collected PiPER Dataset	54.00	506.60	35.82	0.56	15.6	0.5	45.0
Public Robot Data	AgiBot World [39]	285.56	2644.08	160.15	1.16	7.2	0.2	32.53
	RoboMIND [40]	6.56	60.75	10.27	–	–	100.0	0.75
	RoboCOIN [41]	32.00	302.94	44.27	–	–	100.0	3.65
	DROID [42]	27.00	500.82	92.23	–	–	100.0	3.07
Robot-Free UMI Data	Self-Collected UMI Dataset	9.50	90.60	26.02	0.06	2.3	30.8	8.0
Simulated Robot Data	RoboTwin 2.0 [43]	8.16	42.16	35.73	0.145	4.1	26.8	3.94
	LIBERO [44]	0.12	3.86	1.71	0.005	2.9	0.12	0.06
Ego-Centric Human Data	EgoDex [45]	89.24	831.00	338.23	2.96	8.8	18.8	3.0
Total		512.14	4982.81	744.43	4.89	6.6	–	–

space, whereas $\hat{v}_\theta^{a,(e)}$ is integrated directly in the raw action space from $\rho = 1$ (noise) to $\rho = 0$ (data). At each flow step, the current raw action sample is re-encoded by $E_a^{(e)}$ before the next velocity evaluation. In autonomous planning, the model repeatedly updates event and visual memory, infers $\hat{q}_t$, predicts the next chunk, and rolls forward with new observations. In fine-grained instruction following, the externally specified s_t directly steers each atomic action without requiring the model to infer the subgoal from the high-level instruction alone.

3.5 ManipEvent-5M: Event-Based Multimodal Dataset

To facilitate native pretraining of WorldScape Policy 2.0, we construct **ManipEvent-5M**, the event-based multimodal dataset illustrated in Fig. 5. It aggregates human-arm egocentric videos, robot-free UMI demonstrations, simulated trajectories, and real-robot data from self-collected and public sources. Each trajectory is decomposed into an ordered sequence of events:

$$\tau = \{(o_{t_b^j:t_e^j}, a_{t_b^j:t_e^j}, y, c_j, p_{\text{goal}}^j, p_{\text{video}}^j)\}_{j=1}^{N_\tau}, \quad (18)$$

where (t_b^j, t_e^j) is the temporal boundary of event j , y is the episode-level instruction, c_j is its fine-grained event caption, and p_{goal}^j and p_{video}^j are optional visual prompts. This schema separates global task intent from event-level subgoal semantics while preserving their alignment with observations and actions.

Action canonicalization and human-hand retargeting. Robot trajectories are converted to the Cartesian end-effector action representation defined in Sec. 3.1. For human-arm data, estimated hand poses are retargeted to the corresponding robot gripper poses and commands before pretraining, yielding action-aligned human demonstrations under the same position–rotation–grripper semantics. This canonical semantic layout enables joint pretraining across human and robot data, while the embodiment-specific action encoders and decoders preserve distinct kinematic mappings and control conventions.

Text-prompt construction. We convert each trajectory into an episode-level task description and temporally aligned subtask captions using a four-stage hierarchical annotation procedure with Qwen3-VL [46], as illustrated in Fig. 6.

(1) *Two-stage labeling.* We first establish a deterministic temporal structure from robot-state signals and then perform open-vocabulary semantic relabeling with the VLM. For temporal segmentation, embodiment-normalized translational and rotational velocities are computed from end-effector or hand trajectories and smoothed over time. Motion-state transitions, significant gripper-state changes, and episode endpoints form candidate boundaries. Short segments are merged with adjacent events, and an adaptive granularity constraint prevents both semantically overloaded segments and fragments caused by sensor jitter. Each resulting event is also assigned a coarse primitive prior, such as move, grasp/release, or idle, based on motion and gripper signals. Given these fixed temporal boundaries, the VLM determines

the semantic content of the trajectory and each event. This two-stage design decouples *when* an action changes from *what* the action means, reducing missed steps, ordering errors, and temporal misalignment that arise when directly captioning a long video.

(2) *Hierarchical frame sampling.* Captioning is performed at both episode and event levels. At the episode level, we sample up to 16 representative timestamps from the full trajectory and provide their images together with the ordered event list, temporal boundaries, primitive-action priors, and execution outcome. Qwen3-VL returns a structured `high_level` description that summarizes the task actually completed, i.e., a hindsight task description rather than a verbatim copy of the original command. This makes the label robust to missing instructions, execution deviations, and recovery attempts. At the event level, we uniformly sample up to 12 timestamps within each segment and collect their available camera views. Conditioned on the episode-level `high_level`, event boundaries, primitive prior, and local visual evidence, the VLM generates a single-action `step_text`. Separating global summarization from local captioning binds every instruction to a specific event while avoiding omissions and ordering errors in long action sequences.

(3) *Embodied prompt conditioning.* We translate the annotation guidelines into structured prompt constraints rather than requesting an unconstrained video summary. The prompt supplies the temporal metadata and primitive-action prior, and encourages each caption to identify the action order, acting end effector, target object, initial and terminal states, contact mode, motion direction, object interaction, body motion, and, when present, failure or recovery behavior. These constraints increase semantic density while retaining a natural-language action sentence; for example, the caption specifies which gripper interacts with which object and toward which target, rather than producing a generic description such as “move the cup.”

(4) *Multi-view conflict resolution.* For event-level annotation, multi-view observations are presented with a local-first organization, `LEFT_GRIPPER` $\rightarrow$ `RIGHT_GRIPPER` $\rightarrow$ `HEAD`. The head camera provides global scene and task context, whereas wrist/gripper cameras offer more reliable evidence about contact, grasp state, and local target identity. The prompt therefore instructs the VLM to prioritize gripper-view evidence when it conflicts with the head view, mitigating target-object confusion in cluttered scenes. Finally, outputs are parsed and validated as structured records; invalid episode captions trigger a stricter retry or rule-based fallback, while a failed event caption is replaced locally without discarding valid labels from the remaining trajectory.

As summarized in Table 1 and Fig. 7, ManipEvent-5M combines large-scale visual coverage with event-level subtask decomposition across heterogeneous data sources. Beyond fine-grained text annotations, we construct goal-image and video-demonstration prompts to support visual goal specification and cross-embodiment in-context adaptation.

Goal-image prompt construction. We construct two complementary types of goal-image prompts. A *first-view goal image* is directly extracted from the terminal observation of each segmented event, depicting the desired post-condition from the same camera configuration and embodiment as the corresponding action trajectory. It therefore provides an unambiguous visual target for learning how the current state should transition after an atomic action. A *third-view goal image* is sampled from a task-matched human demonstration at the completion of the corresponding event. Such images retain the human demonstrator and show the target object configuration from an external viewpoint. They require the policy to abstract away the demonstrator and viewpoint, infer the intended outcome, and reproduce the demonstrated state using the robot embodiment.

Video-prompt construction. Video prompts are designed primarily for human-to-robot and robot-to-robot task transfer. For human-to-robot transfer, we pair self-collected real-robot trajectories with UMI demonstrations collected for the same tasks. Pairs are matched by the global task, ordered event sequence, and terminal outcome, ensuring that the human and robot demonstrations express consistent action procedures and reducing ambiguity about which skill should be transferred. For robot-to-robot transfer, we collect or retrieve trajectories of the same semantic task executed by different robot embodiments and align them using their ordered event descriptions and goal states. The resulting context-target pairs encourage the model to identify embodiment-invariant object correspondences, action structure, and task intent while adapting execution to the target robot.

Figs. 9 and 10 show representative multimodal prompt samples for desktop cleaning and long-horizon coffee making, respectively. Together, this event-based multimodal organization converts long demonstrations into ordered subgoal segments and aligns language instructions, visual goals, demonstration context, state transitions, and robot actions within a common training interface.

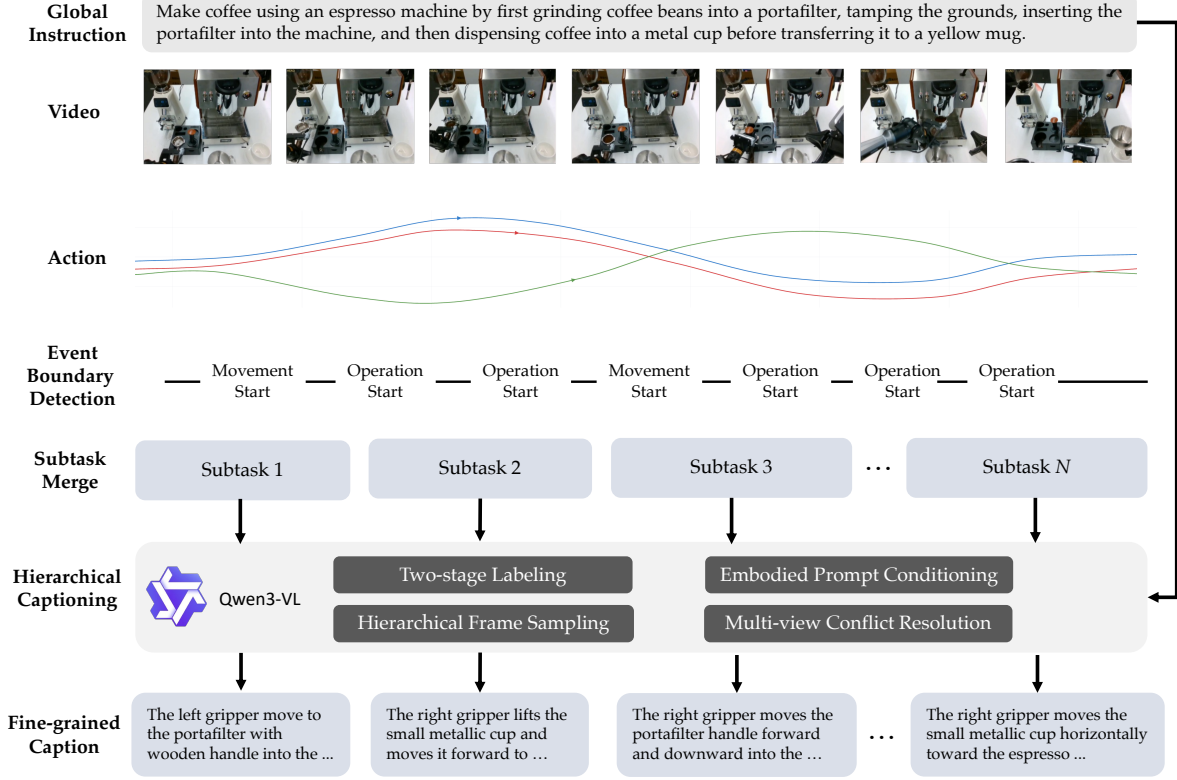


Figure 6: Examples of hierarchical captioning in ManipEvent-5M. The annotation pipeline combines two-stage temporal-semantic labeling, hierarchical episode/event frame sampling, embodied prompt conditioning, and multi-view conflict resolution. The figure shows representative instances that map segmented events to fine-grained captions, complementing the dataset construction pipeline in Fig. 5.

3.6 Training Objectives and Three-Stage Curriculum

WorldScape Policy 2.0 is optimized with a joint video-action objective and, when autonomous planning is enabled, the semantic-forcing objective:

$$\mathcal{L} = \mathcal{L}_{act} + \lambda_w \mathcal{L}_{world} + \lambda_s \mathcal{L}_{sem}. \quad (19)$$

Here, $\mathcal{L}_{act}$ and $\mathcal{L}_{world}$ are flow-matching objectives for action and future visual-latent generation, respectively, and $\mathcal{L}_{sem}$ is defined in Eq. 15; λ_s is set to zero when semantic forcing is inactive. Multimodal conditioning and memory gating are trained end to end through these objectives rather than requiring separate prompt or memory labels. For embodiment e , let $a_0^{(e)}$ denote a clean raw action chunk, whose per-timestep dimension is 20 for the dual-arm setting, and let $a_1^{(e)} \sim \mathcal{N}(0, I)$ be action noise of the same shape. Likewise, z_0 and $z_1 \sim \mathcal{N}(0, I)$ denote the clean and noisy future visual latents. At $\rho \sim \mathcal{U}(0, 1)$, we form $a_\rho^{(e)} = (1 - \rho)a_0^{(e)} + \rho a_1^{(e)}$ and $z_\rho = (1 - \rho)z_0 + \rho z_1$, with raw-space targets $v^{a,(e)} = a_1^{(e)} - a_0^{(e)}$ and $v^z = z_1 - z_0$. Writing $\Xi^{(e)} = \{a_0^{(e)}, a_1^{(e)}, z_0, z_1\}$, the two flow-matching losses are

$$\begin{aligned} \mathcal{L}_{act} &= \mathbb{E}_{e,\rho,\Xi^{(e)}} \left[\|\hat{v}_\theta^{a,(e)}(a_\rho^{(e)}, z_\rho, \rho) - v^{a,(e)}\|_2^2 \right], \\ \mathcal{L}_{world} &= \mathbb{E}_{e,\rho,\Xi^{(e)}} \left[\|\hat{v}_\theta^z(a_\rho^{(e)}, z_\rho, \rho) - v^z\|_2^2 \right]. \end{aligned} \quad (20)$$

Although $a_\rho^{(e)}$ is projected into DiT tokens by $E_a^{(e)}$ for joint video-action attention, the decoded prediction $\hat{v}_\theta^{a,(e)}$ and its target $v^{a,(e)}$ are both defined in the raw action space. Thus, $\mathcal{L}_{act}$ does not impose flow matching on the action embeddings.

Training follows a three-stage curriculum that progressively acquires multimodal controllability, autonomous planning, and downstream interactive manipulation.

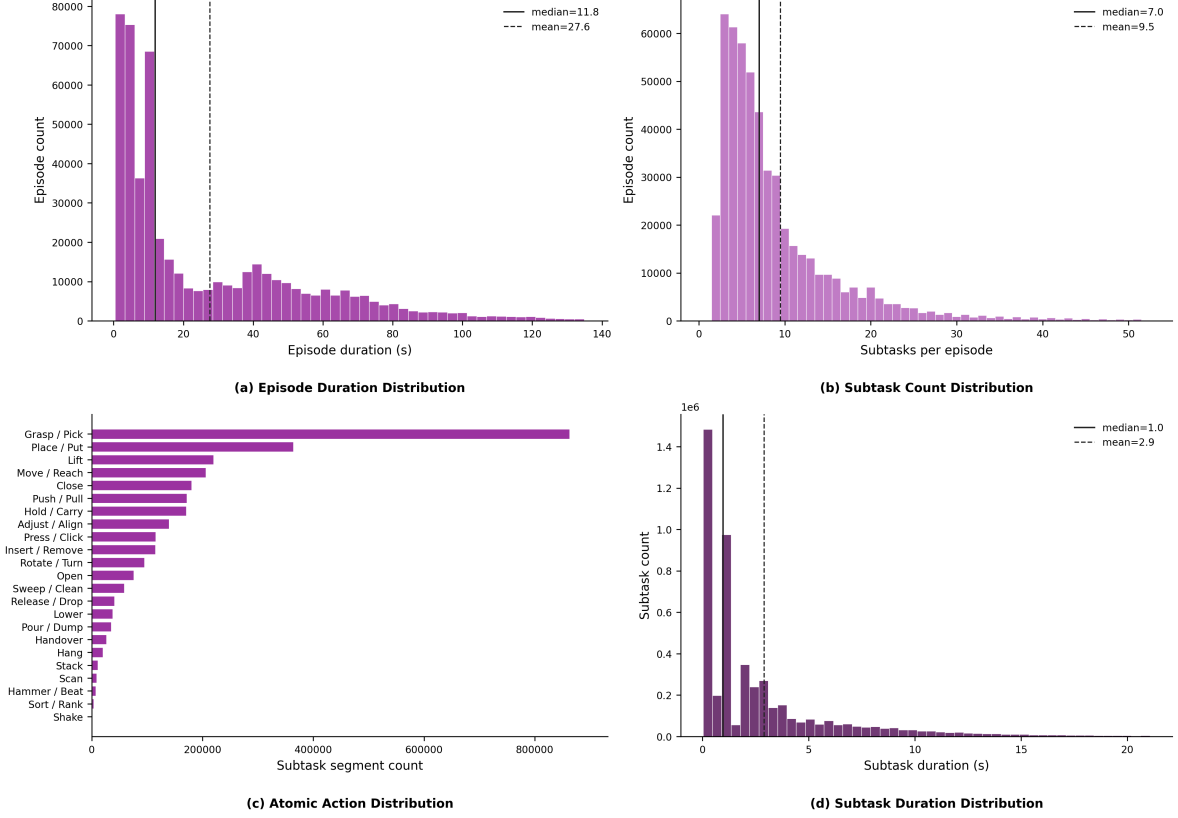


Figure 7: Distribution statistics of ManipEvent-5M. The figure shows the detailed distribution of episode duration, subtask count, atomic actions, and subtask duration across diverse sources from real-world robot data, robot-free data, simulated robot data, and egocentric human data in the constructed ManipEvent-5M.

Stage 1: event-grounded multimodal WAM pretraining. We first pretrain the causal video-action backbone on ManipEvent-5M using event-level samples paired with fine-grained captions, goal images, video demonstrations, visual transitions, and robot actions. Different prompt modalities are sampled as interchangeable control conditions: fine-grained captions are injected through T5 cross-attention, while goal images and video demonstrations are encoded as persistent visual prefill. The short-term causal visual memory is already active at this stage: recent observation chunks are dynamically maintained as clean visual prefill, enabling the DiT to model local temporal continuity through causal self-attention. Stage 1 does not yet introduce the VLM-based event memory or autonomous latent subgoal reasoning. Its objective is

$$\mathcal{L}^{(1)} = \mathcal{L}_{act} + \lambda_w \mathcal{L}_{world}. \quad (21)$$

This stage establishes fine-grained language-video-action grounding, short-horizon visual dynamics, and controllable video-action generation before event-level memory reasoning is introduced.

Stage 2: reasoning-augmented memory mid-training. Starting from the Stage-1 WAM and retaining its short-term causal visual memory, we introduce the VLM planning branch and long short-term event memory for autonomous planning. This stage uses temporally extended robot trajectories, for which episode-level instructions provide global task intent and event-level captions identify the active atomic subgoals. Given only the high-level instruction as input to the planning branch, the model constructs memory-enhanced VLM tokens $\hat{q}_t$ by retrieving global-history, local-active, and event-boundary latents together with the compact history bank; local-active and boundary chunks retain full-token anchor representations for detailed retrieval. The corresponding fine-grained T5 embedding provides a training-time semantic target through $\mathcal{L}_{sem}$, but is not directly used as the autonomous-planning condition. The Stage-2 objective is

$$\mathcal{L}^{(2)} = \mathcal{L}_{act} + \lambda_w \mathcal{L}_{world} + \lambda_s \mathcal{L}_{sem}. \quad (22)$$

By aligning latent subgoal reasoning with the fine-grained semantic space learned in Stage 1, semantic forcing transfers

the pretrained language understanding and controllable video-action generation capabilities to the autonomous planning pathway. The model consequently learns to track task progress, infer the active subgoal, and select the next action without requiring an explicit fine-grained instruction at inference time.

Stage 3: downstream interaction post-training. Finally, we post-train the model on downstream robot tasks using the interaction mode required by each task. Long-horizon manipulation tasks activate high-level instructions, reasoning-augmented event memory, and online visual history for autonomous planning. Fine-grained instruction-following tasks directly condition the WAM on T5 subtask embeddings. In-context adaptation tasks retain goal images or video demonstrations as persistent visual prefill and combine them with online visual memory and, when required, latent reasoning. This stage adapts the shared pretrained model to task-specific embodiments and action distributions while preserving a unified interface for autonomous planning, explicit instruction following, and visual-prompted adaptation.

4 Experiments

4.1 Benchmark Setup and Evaluation Protocol

We organize the evaluation around three capabilities targeted by WorldScape Policy 2.0: simulation generalization, memory-grounded long-horizon planning, and real-world controllability. **Simulation benchmarks** are conducted on RoboTwin 2.0 [43] and LIBERO [44]. RoboTwin 2.0 evaluates manipulation robustness across 50 diverse tasks under clean and randomized settings, with all models fine-tuned for 40K steps and evaluated over 100 trials per task. LIBERO complements this evaluation with language-conditioned manipulation across the Spatial, Object, Goal, and Long suites, testing generalization over layouts, objects, goals, and long-horizon procedures.

Across simulation and real-robot experiments, we report task success, subgoal completion, language following, progress consistency, and in-context adaptation. **Ablation studies** are conducted on RoboTwin 2.0 to isolate memory components, staged training, semantic forcing, and long-term memory length. **Real-robot evaluation** is conducted on an AgileX Robotics dual-arm PiPER platform and covers long-horizon complex manipulation, memory-reasoning and cross-embodiment ICL, and fine-grained instruction following. For each real-world task, we perform 20 trials and report the averaged success rate and task-specific progress score.

4.2 Baselines and Evaluation Settings

We compare WorldScape Policy 2.0 with representative methods from three families. WAM baselines include FastWAM [1] and MemoryWAM [7], which test whether efficient future prediction or persistent memory alone is sufficient for long-horizon manipulation. VLA baselines include OpenVLA [47], RT-X-style generalist policies [48], Qwen-VLA [30], and FineVLA [31], which evaluate language-conditioned and fine-grained control. Low-level imitation baselines include ACT [49] and Diffusion Policy [50]. Each baseline is evaluated under its supported prompt interface, and we separately report text-only, goal-image-conditioned, and video-context-conditioned settings when applicable.

4.3 Implementation Details

Model configurations. For event annotation, we use Qwen3-VL-32B [46] to generate and verify segment-level subtask captions, temporal boundaries, and prompt metadata. The reasoning-augmented LSTM uses a lightweight Qwen3-VL-4B model [46] as the latent reasoning backbone. To reduce memory cost, this module receives only the head-view observation image, resized to 320×160 .

The WAM input is constructed by concatenating three camera views into a single visual canvas. The video DiT is initialized from the Wan2.2-5B text-to-video pretrained model [51], and the video DiT and action DiT share the same backbone design. Video-action interaction is implemented with joint bidirectional attention between video and action tokens, while attention to clean tokens and visual-history tokens remains causal to preserve autoregressive prediction. During multi-embodiment pretraining, each embodiment uses its own action encoder-decoder pair $(E_a^{(e)}, D_a^{(e)})$, while the video-action backbone is shared. The short-term visual history contains four chunks. The long-term memory keeps eight history chunks by default, and this length can be adjusted according to task horizon and memory demand.

Table 2: Evaluation on the RoboTwin 2.0 benchmark under different settings (Clean vs. Randomized, among 50 tasks). See Table 10 for our full evaluation results.

Model	Clean	Random	Average
# Vision-Language-Action (VLA) Models			
π_0	65.9%	58.4%	62.2%
X-VLA	72.9%	72.8%	72.9%
$\pi_{0.5}$	82.7%	76.8%	79.8%
StarVLA	88.2%	88.3%	88.3%
Abot-M0	81.2%	80.4%	80.8%
LingBot-VLA	86.5%	85.3%	85.9%
HoloBrain-0-QW	91.9%	92.3%	92.1%
# World Action Models			
JEPA-VLA	73.5%	-	73.5%
Motus	88.7%	87.0%	87.9%
GigaWorld-Policy	85.6%	85.3%	85.5%
LingBot-VA	92.9%	91.6%	92.3%
Fast-WAM	91.9%	91.8%	91.9%
Being-H0.7	90.2%	89.6%	89.9%
WorldScape Policy 2.0	-%	-%	-%

Table 3: Evaluation on the LIBERO and LIBERO-Plus benchmarks. For LIBERO-Plus, models trained on LIBERO are evaluated in a zero-shot manner without further fine-tuning across seven distribution shifts: camera, robot, language, lighting, background, noise, and layout.

Methods	Size	LIBERO					LIBERO-Plus							
		Spatial	Object	Goal	Long	Avg	Cam	Robot	Lang	Light	BG	Noise	Layout	Avg
MemoryVLA	7B	98.4	98.4	96.4	93.4	96.7	-	-	-	-	-	-	-	-
Octo	0.1B	78.9	85.7	84.6	51.1	75.1	-	-	-	-	-	-	-	-
FLOWER	1B	97.1	96.7	95.6	93.5	95.7	-	-	-	-	-	-	-	-
SmoIVLA	2B	93.0	94.0	91.0	77.0	88.8	-	-	-	-	-	-	-	-
GROOT-N1	3B	94.4	97.6	93.0	90.6	93.9	-	-	-	-	-	-	-	-
OpenVLA	7B	84.7	88.4	79.2	53.7	76.5	0.8	3.5	23.0	8.1	34.8	15.2	28.5	15.6
OpenVLA-OFT	7B	97.6	98.4	97.9	94.5	97.1	56.4	31.9	79.5	88.7	93.3	75.8	74.2	69.6
WorldVLA	1B	85.6	89.0	82.6	59.0	79.1	0.1	27.9	41.6	43.7	17.1	10.9	38.0	25.0
UniVLA	7B	96.5	96.8	95.6	92.0	95.2	1.8	46.2	69.6	69.0	81.0	21.2	31.9	42.9
π_0	3B	96.8	98.8	95.8	85.2	94.1	13.8	6.0	58.8	85.0	81.4	79.0	68.9	53.6
π_0 -FAST	3B	96.4	96.8	88.6	60.2	85.5	65.1	21.6	61.0	73.2	73.2	74.4	68.8	61.6
$\pi_{0.5}$	3B	98.8	98.2	98.0	92.4	96.9	-	-	-	-	-	-	-	-
X-VLA	0.9B	98.2	98.6	97.8	97.6	98.1	22.2	87.8	73.1	88.2	95.3	61.8	70.7	69.7
WorldScape Policy 2.0	-	-	-	-	-	-	-	-	-	-	-	-	-	-

Training setup. The action interface follows Sec. 3.1: each arm contributes 3D position, a continuous 6D rotation representation, and a one-DoF gripper command, giving a 20D raw action for dual-arm control. As specified in Eq. 20, action flow matching is optimized directly in this raw space; the action encoder and decoder serve only as embodiment-specific interfaces to the shared DiT. Event-grounded pretraining uses batch size 768 and learning rate 5×10^{-4} ; post-training uses batch size 128 and learning rate 6×10^{-5} for 50K steps.

4.4 Simulation Benchmark Results

The simulation benchmark evaluates zero-shot manipulation and cross-task generalization on RoboTwin 2.0 [43] and LIBERO [44]. On RoboTwin 2.0, we select tasks with compositional object arrangements, visually similar intermediate states, and long-horizon dependencies. These tasks stress progress-aware planning because the correct action depends not only on the current image but also on which subtasks have already been completed. On LIBERO, we evaluate language-conditioned manipulation across task suites that require transferring object, spatial, and procedural knowledge.

Table 2 reports RoboTwin 2.0 results under clean and randomized settings, measuring whether WorldScape Policy 2.0 improves manipulation robustness across diverse tasks. Table 3 reports LIBERO results in terms of task success, language-following accuracy, and cross-task generalization. Together, the two benchmarks separate three factors: whether world-action modeling improves over pure imitation policies, whether long short-term memory improves

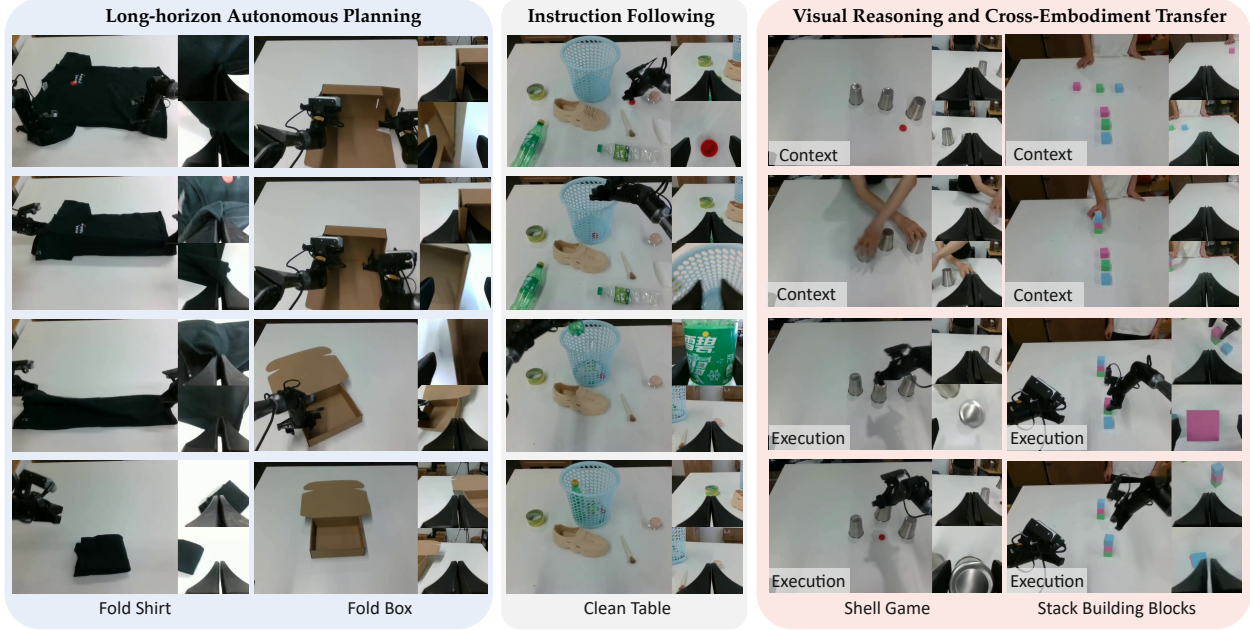


Figure 8: Illustration of real-world tasks on the dual-arm PiPER platform. The benchmark covers long-horizon autonomous planning through shirt and box folding, fine-grained instruction following through table cleaning, and visual-context reasoning and cross-embodiment transfer through the shell game and demonstration-conditioned block stacking, respectively.

Table 4: Real-robot manipulation benchmark across diverse capabilities. The evaluation reports episode-level task success and progress for autonomous planning, sequential instruction following, and in-context reasoning / skill transfer with text, goal-image, and video-demonstration prompts.

Capability	Tasks	Prompt / Context	$\pi_{0.5}$	DreamZero	WorldScape Policy 2.0
Long-Horizon Autonomous Planning	Folding Clothes	Global Instruction	—	—	—
	Folding Box		—	—	—
Memory-Dependent Visual Reasoning	Shell Game	Video Demo	—	—	—
Cross-Embodiment Skill Transfer	Stacking Blocks	Goal Image	—	—	—
		Video Demo	—	—	—
Sequential Fine-Grained Control	Clean Table (Full Sequence)	Sequential Subtask Captions	—	—	—

over bounded-context WAMs, and whether multimodal prompts provide additional controllability beyond text-only conditioning.

4.5 Real-World Evaluation Results

We conduct real-world experiments on the dual-arm PiPER platform, evaluating capabilities that are difficult to fully capture in simulation, with the five real-world tasks illustrated in Fig. 8. **Long-horizon complex tasks** test whether WorldScape Policy 2.0 can track progress and complete multi-stage manipulation from coarse instructions, such as folding clothes or making coffee. **In-context learning** includes memory-reasoning tasks such as shell-game ball guessing, where the robot must use observed history to infer a hidden state, and **cross-embodiment transfer**, where the robot conditions on human-to-robot or robot-to-robot demonstration videos. **Fine-grained instruction following** evaluates whether event-level language prompts enable precise control over atomic actions and staged object arrangements. Table 4 reports episode-level success and progress across capabilities; its table-cleaning row evaluates the full sequence driven by successive subtask captions. Table 5 instead isolates text grounding by evaluating each atomic

Table 5: Fine-grained text-instruction following on the real-world table-cleaning task. In-domain object categories are included in task-specific post-training, whereas out-of-domain categories are held out and introduced only at evaluation. Each row independently evaluates one atomic instruction under a reset scene. We report instruction success rate over 20 trials.

Task	Domain Split	Fine-Grained Text Instruction	$\pi_{0.5}$	DreamZero	WorldScape Policy 2.0
Clean Table	In-domain	Pick up the <i>green bottle</i> and place it into the <i>basket</i> .	—	—	—
		Pick up the <i>clear bottle</i> and place it into the <i>basket</i> .	—	—	—
		Pick up the <i>red cup</i> and place it into the <i>basket</i> .	—	—	—
	Out-of-domain	Pick up the <i>shoe</i> and place it into the <i>basket</i> .	—	—	—
		Pick up the <i>tape roll</i> and place it into the <i>basket</i> .	—	—	—
		Pick up the <i>brush</i> and place it into the <i>basket</i> .	—	—	—
In-Domain Average			—	—	—
Out-of-Domain Average			—	—	—
Overall Average			—	—	—

instruction independently, with separate in-domain and held-out object categories. The split-wise instruction success rates distinguish language control on familiar objects from compositional generalization to novel object references, while preventing full-task execution errors from obscuring atomic grounding ability.

4.6 Ablation Study

We first isolate the memory design in Table 6. The no-memory variant uses only the current observation and prompt, testing how far a causal WAM can go without persistent task history. The short-term-only variant retains recent visual-action context and evaluates local action precision. The long-term-only variant stores event-level progress memory and evaluates long-horizon consistency. The long short-term variant combines both memory streams, testing whether local dynamics and global task progress are complementary. The full WorldScape Policy 2.0 model further adds latent reasoning memory for implicit subgoal planning when the next action is not directly specified by the prompt.

We further vary the long-term memory history length in Table 7. Shorter memory reduces computation but may miss distant subgoal evidence, while longer memory covers more completed events at higher memory cost. We use eight chunks as the default setting because it balances long-horizon progress tracking and efficiency, and adjust the length only for tasks that require substantially longer history.

The second ablation in Table 8 evaluates the first two stages of the curriculum and semantic forcing, while applying the same Stage-3 downstream post-training to all variants. The from-scratch variant tests whether WorldScape Policy 2.0 can acquire controllable WAM behavior without ManipEvent-5M pretraining. Stage-1-only training evaluates event-grounded multimodal prompt alignment before memory mid-training. Stage 1 plus Stage 2 without semantic forcing measures whether memory training alone is sufficient for progress-aware prediction. Stage 2 without Stage 1 tests whether memory learning remains effective when the model lacks event-grounded prompt alignment. The full model combines Stage-1 event-grounded pretraining, Stage-2 memory mid-training with semantic forcing, and Stage-3 task-specific adaptation.

Table 9 isolates whether the VLM backbone should remain frozen during Stage-2 memory mid-training. In the frozen setting, the VLM parameters are fixed while the event-memory, retrieval, gating, and projection modules remain trainable. In the jointly fine-tuned setting, the VLM backbone is updated together with these modules. Both variants use the same Stage-1 initialization, Stage-2 data and optimization schedule, and Stage-3 post-training protocol, so the comparison attributes performance differences specifically to VLM updating.

4.7 Qualitative Analysis, Failure Cases, and Summary

Qualitative analysis focuses on three representative behaviors. First, WorldScape Policy 2.0 distinguishes visually similar states using long-term progress memory, for example avoiding repeated actions after a subgoal has already been completed. Second, it adjusts atomic actions according to fine-grained text prompts, showing that event-level

Table 6: Component analysis on RoboTwin 2.0. STM denotes short-term visual memory, LTM denotes long-term event memory, and LSR denotes latent subgoal reasoning.

STM	LTM	LSR	Clean	Random	Average
✗	✗	✗	–	–	–
✓	✗	✗	–	–	–
✓	✓	✗	–	–	–
✓	✓	✓	–	–	–

Table 7: Effect of the long short-term memory length on RoboTwin 2.0. Official setting is adopted and the number of history video chunks are measured.

Event Memory	Visual Memory	Clean	Random	Average
8	2	–	–	–
8	4	–	–	–
8	8	–	–	–
16	4	–	–	–

language conditions affect execution rather than only task selection. Third, it follows goal-image or video-context prompts to solve unseen task variations, especially in memory-reasoning and cross-embodiment transfer settings. We summarize results across simulation benchmarks, RoboTwin 2.0 ablations, and real-robot evaluations by analyzing how each memory component contributes, which training stage is most important for each capability, how semantic forcing affects subgoal reasoning, and where real-robot deployment remains challenging. Current limitations include high-precision contact-rich manipulation, memory compression over extremely long horizons, sensitivity to visual prompt viewpoint changes, and dependence on ManipEvent-5M annotation quality.

5 Conclusion

We introduced WorldScape Policy 2.0, a controllable World Action Model that augments causal video-action prediction with reasoning-aware long short-term memory. By combining short-term visual-action context, long-term event-level progress memory, and latent subgoal reasoning, WorldScape Policy 2.0 supports autonomous planning, fine-grained text control, goal-image conditioning, and demonstration-conditioned in-context adaptation within a unified WAM interface. Together with ManipEvent-5M, an event-grounded multimodal pretraining dataset aligning visual transitions, actions, subtask captions, goal images, and video contexts, WorldScape Policy 2.0 reframes WAM pretraining from passive future prediction into controllable and memory-grounded world-action modeling. Experiments and ablations on RoboTwin 2.0, LIBERO, and the dual-arm PiPER platform evaluate the roles of memory, staged training, semantic forcing, multimodal prompting, and real-robot transfer, while remaining challenges include contact-rich precision, very long-horizon memory compression, viewpoint-robust visual prompting, and scalable event annotation.

Table 8: Ablation of Stage-1 event-grounded pretraining and Stage-2 memory mid-training on RoboTwin 2.0. SF denotes semantic forcing. All variants use the same Stage-3 downstream post-training protocol.

Stage-1	Stage-2	SF	Clean	Random	Average
✗	✗	✗	—	—	—
✓	✗	✗	—	—	—
✓	✓	✗	—	—	—
✗	✓	✓	—	—	—

Table 9: Effect of VLM freezing during Stage-2 memory mid-training on RoboTwin 2.0. All other trainable modules, data, and optimization settings are held fixed.

VLM Status	Clean	Random	Average
Frozen	—	—	—
Jointly Fine-tuned	—	—	—

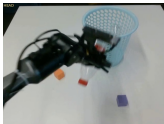
Global Instruction	Sequentially pick up two plastic bottles and two colored cubes from the tabletop and place them into a light blue perforated basket.					
Current Observation						...
Goal Image Condition						...
X-Emb. Video Demonstration						...
Fine-Grained Sub-task Caption	Left gripper grasps the transparent plastic bottle with a green label from the tabletop and lifts it upward toward the light blue basket.	Left gripper closes around the transparent plastic bottle with a red cap and label, lifting it from the table surface.	Left gripper lifts the transparent plastic bottle with a red cap from the table then drop it to blue basket.	Left gripper closes around the orange cube on the table, picks it up then move it to light blue basket.	Left gripper grasps the purple cube on the table, moves it toward the light blue basket and releases it inside.	...

Figure 9: Multimodal prompt example from ManipEvent-5M: desktop cleaning. The sample pairs a global task instruction and current observations with goal-image conditioning, event-level fine-grained subtask captions, and a cross-embodiment video demonstration for placing bottles and colored cubes into a basket.

Global Instruction	Make coffee using an espresso machine by first grinding coffee beans into a portafilter, tamping the grounds, inserting the portafilter into the machine, and then dispensing coffee into a metal cup before transferring it to a yellow mug.				
Current Observation					 ...
Goal Image Condition					 ...
X-Emb. Video Demonstration					 ...
Fine-Grained Sub-task Caption	Left gripper lifts the metallic portafilter with wooden handle from the black holder.	Left gripper holds the portafilter steady under the coffee grinder as ground coffee falls into it.	Left gripper inserts the metal portafilter into the group head of the coffee machine.	Left gripper grasps the wooden-topped tamper from the black holder and presses it downward into the portafilter to compact the coffee grounds.	Left gripper grasps the wooden-handled portafilter and inserts it into the espresso machine's group head in a forward motion. ...

Figure 10: Multimodal prompt example from ManipEvent-5M: long-horizon coffee making. The sample illustrates how a multi-stage task is represented by a global instruction, current observations, a goal-image condition, fine-grained subtask captions, and a cross-embodiment video demonstration.

Table 10: Detailed evaluation results (per-task success rates) on the RoboTwin 2.0 benchmark (Clean vs Randomized, 50 tasks).

Simulation Task	$\pi_{0.5}$		GigaWorld-Policy		Motus		LingBot-VA		Fast-WAM		WorldScape Policy 2.0	
	Clean	Rand.	Clean	Rand.	Clean	Rand.	Clean	Rand.	Clean	Rand.	Clean	Rand.
Adjust Bottle	100	99	100	100	89	93	90	94	100	100	100	100
Beat Block Hammer	96	93	86	86	95	88	96	98	99	97	100	99
Blocks Ranking RGB	92	85	92	96	92	91	99	98	100	100	100	100
Blocks Ranking Size	49	26	44	48	75	63	94	96	94	98	96	98
Click Alarmclock	98	89	100	100	100	100	99	100	100	100	100	100
Click Bell	99	66	100	100	100	100	100	100	100	100	100	100
Dump Bin Bigbin	92	97	92	100	95	91	89	96	97	96	98	96
Grab Roller	100	100	100	100	100	100	100	100	100	100	100	100
Handover Block	66	57	80	80	86	73	99	78	95	81	99	96
Handover Mic	98	97	72	72	78	63	94	96	99	100	99	98
Hanging Mug	18	17	16	12	38	38	40	28	58	62	82	83
Lift Pot	96	85	98	98	96	99	100	99	100	100	100	100
Move Can Pot	51	55	76	78	34	74	96	97	90	88	97	96
Move Pillbottle Pad	84	61	90	90	93	96	99	99	100	99	100	100
Move Playingcard Away	96	84	78	72	100	96	100	99	100	100	100	100
Move Stapler Pad	56	42	92	82	83	85	91	79	77	64	94	92
Open Laptop	90	96	96	98	95	91	92	94	98	100	100	100
Open Microwave	34	77	74	66	95	91	82	86	62	45	45	62
Pick Diverse Bottles	81	71	82	70	90	91	89	82	80	85	97	99
Pick Dual Bottles	93	63	86	86	96	90	100	99	100	96	100	100
Place A2B Left	87	82	94	88	88	79	97	93	95	93	100	99
Place A2B Right	87	84	90	92	91	87	97	95	93	99	99	100
Place Bread Basket	77	64	82	82	91	94	97	95	91	93	97	96
Place Bread Skillet	85	66	94	90	96	83	95	90	90	93	95	95
Place Burger Fries	94	87	98	96	98	98	97	95	96	99	98	99
Place Can Basket	62	62	78	74	81	76	81	84	71	69	85	87
Place Cans Plasticbox	94	84	100	100	98	94	100	99	99	96	100	100
Place Container Plate	99	95	98	96	98	99	99	97	96	98	100	100
Place Dual Shoes	75	75	96	84	93	87	94	89	94	88	96	94
Place Empty Cup	100	99	90	90	99	98	100	100	100	100	100	99
Place Fan	87	85	92	94	91	87	99	93	96	96	100	98
Place Mouse Pad	60	39	88	90	66	68	93	96	83	89	94	92
Place Object Basket	80	76	90	92	81	87	91	88	89	88	94	95
Place Object Scale	86	80	88	80	88	85	96	95	90	97	97	94
Place Object Stand	91	85	100	98	98	97	99	96	90	94	98	99
Place Phone Stand	81	81	82	72	87	86	97	97	97	99	100	100
Place Shoe	92	93	98	96	99	97	98	98	96	99	99	99
Press Stapler	87	83	96	96	93	98	85	82	90	97	96	97
Put Bottles Dustbin	84	79	72	70	81	79	87	91	95	90	97	98
Put Object Cabinet	80	79	74	74	88	71	85	87	94	89	91	89
Rotate QRcode	89	87	90	84	89	73	96	91	93	89	95	93
Scan Object	72	65	60	64	67	66	96	91	89	92	95	96
Shake Bottle	99	97	100	100	100	97	100	97	100	100	100	100
Shake Bottle Horizontally	99	99	100	98	100	98	100	99	100	100	100	100
Stack Blocks Three	91	76	70	78	91	95	99	98	95	97	100	100
Stack Blocks Two	97	100	100	94	100	98	100	99	100	100	100	100
Stack Bowls Three	77	71	70	72	79	87	86	83	80	81	87	89
Stack Bowls Two	95	96	96	92	98	98	94	98	92	98	99	98
Stamp Seal	79	55	96	98	93	92	96	97	90	94	100	99
Turn Switch	62	54	82	84	84	78	44	45	61	59	83	84
Average (%)	82.74	76.76	85.62	85.28	88.66	87.02	92.90	91.50	91.88	91.78	96.04	96.16

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